

## Moller scattering 2

For transition amplitudes  $\mathcal{M}_t$  we have

$$\mathcal{M}_t = \iint \underbrace{\bar{\psi}_4(y)(-ie\gamma^\mu)\psi_2(y)}_{\text{vertex}} \underbrace{G(x,y)}_{\substack{\text{photon} \\ \text{propagator}}} \underbrace{\bar{\psi}_3(x)(-ie\gamma_\mu)\psi_1(x)}_{\text{vertex}} d^4x d^4y$$

where  $G(x, y)$  is the photon propagator

$$G(x, y) = \frac{1}{(2\pi)^4} \int \frac{\exp(-ip(x-y))}{p^2 + i\epsilon} d^4p$$

Moller scattering wave functions normalized as  $\langle\psi|\psi\rangle = 1$  over volume  $V$ .

$$\begin{aligned}\psi_1(x^\mu) &= \sqrt{2EV} u_1 \exp(-ip_1^\mu g_{\mu\nu} x^\nu) \\ \psi_2(x^\mu) &= \sqrt{2EV} u_2 \exp(-ip_2^\mu g_{\mu\nu} x^\nu) \\ \psi_3(x^\mu) &= \sqrt{2EV} u_3 \exp(-ip_3^\mu g_{\mu\nu} x^\nu) \\ \psi_4(x^\mu) &= \sqrt{2EV} u_4 \exp(-ip_4^\mu g_{\mu\nu} x^\nu)\end{aligned}$$

For clarity we abbreviate as

$$\begin{aligned}\psi_1(x) &= \sqrt{2EV} u_1 \exp(-ip_1 x) \\ \psi_2(x) &= \sqrt{2EV} u_2 \exp(-ip_2 x) \\ \psi_3(x) &= \sqrt{2EV} u_3 \exp(-ip_3 x) \\ \psi_4(x) &= \sqrt{2EV} u_4 \exp(-ip_4 x)\end{aligned}$$

Substitute for the wave functions in  $\mathcal{M}_t$  and factor the spinors.

$$\begin{aligned}\mathcal{M}_t &= -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) \\ &\quad \times \iint \exp(ip_4 y) \exp(-ip_2 y) G(x, y) \exp(ip_3 x) \exp(-ip_1 x) d^4x d^4y\end{aligned}$$

Substitute for  $G(x, y)$ .

$$\begin{aligned}\mathcal{M}_t &= -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) \\ &\quad \times \frac{1}{(2\pi)^4} \iiint \exp(ip_4 y) \exp(-ip_2 y) \frac{\exp(-ip(x-y))}{p^2 + i\epsilon} \exp(ip_3 x) \exp(-ip_1 x) d^4x d^4y d^4p\end{aligned}$$

Combine exponentials.

$$\begin{aligned}\mathcal{M}_t &= -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) \\ &\quad \times \frac{1}{(2\pi)^4} \iiint \frac{\exp(i(p_4 - p_2 + p)y) \exp(i(p_3 - p_1 - p)x)}{p^2 + i\epsilon} d^4x d^4y d^4p\end{aligned}$$

The integrals over  $x$  and  $y$  are solved as

$$\begin{aligned}\int \exp(i(p_3 - p_1 - p)x) d^4x &= (2\pi)^4 \delta^4(p_3 - p_1 - p) \\ \int \exp(i(p_4 - p_2 + p)y) d^4y &= (2\pi)^4 \delta^4(p_4 - p_2 + p)\end{aligned}$$

Hence

$$\mathcal{M}_t = -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) (2\pi)^4 \int \frac{\delta^4(p_4 - p_2 + p) \delta^4(p_3 - p_1 - p)}{p^2 + i\epsilon} d^4p$$

The integral over  $p$  vanishes except for

$$p = p_3 - p_1 = p_2 - p_4$$

Hence

$$\mathcal{M}_t = -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) (2\pi)^4 \frac{\delta^4(p_4 + p_3 - p_2 - p_1)}{(p_3 - p_1)^2}$$

For momentum vectors

$$\begin{aligned}p_1 &= \begin{pmatrix} E \\ 0 \\ 0 \\ p \end{pmatrix}_{e^- \rightarrow} & p_2 &= \begin{pmatrix} E \\ 0 \\ 0 \\ -p \end{pmatrix}_{\leftarrow e^-} & p_3 &= \begin{pmatrix} E \\ p \sin \theta \cos \phi \\ p \sin \theta \sin \phi \\ p \cos \theta \end{pmatrix}_{e^- \nearrow} & p_4 &= \begin{pmatrix} E \\ -p \sin \theta \cos \phi \\ -p \sin \theta \sin \phi \\ -p \cos \theta \end{pmatrix}_{\nwarrow e^-}\end{aligned}$$

we have  $\delta^4(p_4 + p_3 - p_2 - p_1) = \delta^4(0) = 1$  hence

$$\mathcal{M}_t = -\frac{e^2}{(2EV)^2} (\bar{u}_4 \gamma^\mu u_2) (\bar{u}_3 \gamma_\mu u_1) \frac{(2\pi)^4}{t}$$

where  $t$  is the Mandelstam variable

$$t = (p_1 - p_3)^2$$

Transition amplitudes  $\mathcal{M}_u$  have  $\psi_3$  and  $\psi_4$  interchanged relative to  $\mathcal{M}_t$ .

$$\mathcal{M}_u = \iint \bar{\psi}_3(y) (-ie\gamma^\mu) \psi_2(y) G(x, y) \bar{\psi}_4(x) (-ie\gamma_\mu) \psi_1(x) d^4x d^4y$$

Hence for  $\mathcal{M}_u$  we have

$$\mathcal{M}_u = -\frac{e^2}{(2EV)^2} (\bar{u}_3 \gamma^\mu u_2) (\bar{u}_4 \gamma_\mu u_1) \frac{(2\pi)^4}{u}$$

where  $u$  is the Mandelstam variable

$$u = (p_1 - p_4)^2$$